

Hw5-NoSQL

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Q1

```
> db.airbnb_listings.countDocuments({})  
< 5555
```

Q2

```
> db.airbnb_listings.countDocuments({"property_type": "House"})  
< 606
```

Q3

```
> db.airbnb_listings.aggregate([  
  { "$group": { "_id": "$property_type", "count": { "$sum": 1 } } },  
  { "$sort": { "count": -1 } }  
)  
< {  
  _id: 'Apartment',  
  count: 3626  
}  
{  
  _id: 'House',  
  count: 606  
}  
{  
  _id: 'Condominium',  
  count: 399  
}  
{  
  _id: 'Serviced apartment',  
  count: 185  
}  
{  
  _id: 'Loft',  
  count: 142  
}
```

```
  _id: 'Lott',  
  count: 142  
}  
{  
  _id: 'Townhouse',  
  count: 108  
}  
{  
  _id: 'Guest suite',  
  count: 81  
}  
{  
  _id: 'Bed and breakfast',  
  count: 69  
}  
{  
  _id: 'Boutique hotel',  
  count: 53  
}  
{  
  _id: 'Guesthouse',  
  count: 50  
}
```

```
{
  _id: 'Hostel',
  count: 34
}
{
  _id: 'Villa',
  count: 32
}
{
  _id: 'Hotel',
  count: 26
}
{
  _id: 'Aparthotel',
  count: 23
}
{
  _id: 'Cottage',
  count: 20
}
{
  _id: 'Other',
  count: 18
}
```

```
{
  _id: 'Cabin',
  count: 15
}
{
  _id: 'Bungalow',
  count: 14
}
{
  _id: 'Resort',
  count: 11
}
{
  _id: 'Farm stay',
  count: 9
}
```

Q4

```
> db.airbnb_listings.countDocuments({ review_scores_cleanliness: { $lt: 5 } })
< 0
```

Q5

```

> db.airbnb_listings.aggregate([
  { "$group": { "_id": ["$room_type", "$cancellation_policy"], "count": { "$sum": 1 } } },
  { "$sort": { "count": 1 } }
])
< {
  _id: [
    'Private room',
    'super_strict_60'
  ],
  count: 6
}
{
  _id: [
    'Shared room',
    'moderate'
  ],
  count: 8
}
{
  _id: [
    'Shared room',
    'strict_14_with_grace_period'
  ],

```

```

    _id: [
      'Shared room',
      'strict_14_with_grace_period'
    ],
    count: 24
  }
  {
    _id: [
      'Entire home/apt',
      'super_strict_30'
    ],
    count: 38
  }
  {
    _id: [
      'Shared room',
      'flexible'
    ],
    count: 51
  }
}

```

```
{
  _id: [
    'Entire home/apt',
    'super_strict_60'
  ],
  count: 73
}
{
  _id: [
    'Private room',
    'moderate'
  ],
  count: 435
}
{
  _id: [
    'Private room',
    'strict_14_with_grace_period'
  ],
  count: 698
}
```

```
{
  _id: [
    'Entire home/apt',
    'flexible'
  ],
  count: 787
}
{
  _id: [
    'Private room',
    'flexible'
  ],
  count: 844
}
{
  _id: [
    'Entire home/apt',
    'moderate'
  ],
  count: 893
}
}

{
  _id: [
    'Entire home/apt',
    'strict_14_with_grace_period'
  ],
  count: 1698
}
```

Q6

```
> db.airbnb_listings.aggregate( [
  { $match: { property_type: "Apartment" } } ,
  { $group: {
    _id: {RoomType: "$room_type", Policy: "$cancellation_policy"},
    count: { $sum: 1 }    }  } ] ).sort({ "_id.Policy" : 1 })
```

```
< {
  _id: {
    RoomType: 'Private room',
    Policy: 'flexible'
  },
  count: 550
}
{
  _id: {
    RoomType: 'Entire home/apt',
    Policy: 'flexible'
  },
  count: 567
}
```

```
{
  _id: {
    RoomType: 'Shared room',
    Policy: 'flexible'
  },
  count: 24
}
{
  _id: {
    RoomType: 'Entire home/apt',
    Policy: 'moderate'
  },
  count: 631
}
{
  _id: {
    RoomType: 'Shared room',
    Policy: 'moderate'
  },
  count: 3
}
```



```
{
  _id: {
    RoomType: 'Private room',
    Policy: 'moderate'
  },
  count: 287
}
{
  _id: {
    RoomType: 'Entire home/apt',
    Policy: 'strict_14_with_grace_period'
  },
  count: 1112
}
{
  _id: {
    RoomType: 'Shared room',
    Policy: 'strict_14_with_grace_period'
  },
  count: 12
}
{
  _id: {
```

```

{
  _id: {
    RoomType: 'Private room',
    Policy: 'strict_14_with_grace_period'
  },
  count: 416
}
{
  _id: {
    RoomType: 'Entire home/apt',
    Policy: 'super_strict_30'
  },
  count: 13
}
{
  _id: {
    RoomType: 'Entire home/apt',
    Policy: 'super_strict_60'
  },
  count: 10
}
{
  }
{
  _id: {
    RoomType: 'Private room',
    Policy: 'super_strict_60'
  },
  count: 1
}

```

Q7

```
> db.airbnb_listings.find(
  { "address.market": "New York", monthly_price: { $gt: 5000 } },
  { _id: 1, monthly_price: 1 }
).sort({ monthly_price: -1 })
< {
  _id: '846854',
  monthly_price: Decimal128('17000.00')
}
{
  _id: '256328',
  monthly_price: Decimal128('12000.00')
}
{
  _id: '598612',
  monthly_price: Decimal128('10000.00')
}
{
  _id: '1897001',
  monthly_price: Decimal128('9495.00')
}
```

```
{
  _id: '6064471',
  monthly_price: Decimal128('8820.00')
}
{
  _id: '640813',
  monthly_price: Decimal128('8000.00')
}
{
  _id: '1146653',
  monthly_price: Decimal128('7500.00')
}
{
  _id: '1155475',
  monthly_price: Decimal128('6999.00')
}
{
  _id: '2253500',
  monthly_price: Decimal128('6250.00')
}

{
  _id: '102995',
  monthly_price: Decimal128('6000.00')
}
```

Q8

```
> db.airbnb_listings.find(
  {},
  { name: 1, property_type: 1, cleaning_fee:1}
).sort({ cleaning_fee: -1 }).limit(10)
< {
  _id: '13927230',
  name: 'Casa completa p olimpíadas com serviços incluído',
  property_type: 'House',
  cleaning_fee: Decimal128('2000.00')
}
{
  _id: '5725151',
  name: 'service apartment with terrace',
  property_type: 'Apartment',
  cleaning_fee: Decimal128('1200.00')
}
{
  _id: '6147746',
  name: 'Stunning Waterfront Marina bay house in Sai Kung',
  property_type: 'House',
  cleaning_fee: Decimal128('1000.00')
}
```

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```
{
  _id: '28884716',
  name: '两房一厅,出租单人床房间(该房间不足99尺,房客不能带超过2个行李箱.另外,双人床房已有1人长住)',
  property_type: 'Serviced apartment',
  cleaning_fee: Decimal128('1000.00')
}
{
  _id: '1176693',
  name: 'BEST REVIEWS*BEST MALLS*SAFE STAY*DIMSUM*CWB*MTR',
  property_type: 'Apartment',
  cleaning_fee: Decimal128('942.00')
}
{
  _id: '20362690',
  name: 'WORLD CLASS MALLS*LUXURY 3BED2BATH*CLEAN*MTR*SAFE',
  property_type: 'Apartment',
  cleaning_fee: Decimal128('942.00')
}
{
  _id: '15488401',
  name: 'INSTAGRAM HOME *MTR*ELEMENT MALL*NEW TRAIN KOWLOON',
  property_type: 'Apartment',
  cleaning_fee: Decimal128('942.00')
```

```
}
{
  _id: '16215566',
  name: "Kahala Ali'i",
  property_type: 'House',
  cleaning_fee: Decimal128('910.00')
}
{
  _id: '5640127',
  name: 'Laulea Kailani Villa (Kauai), Peaceful Luxury',
  property_type: 'House',
  cleaning_fee: Decimal128('850.00')
}
{
  _id: '25065550',
  name: "Lanikai La'i Hale with Oceanview 3BD plus Loft/3BA",
  property_type: 'House',
  cleaning_fee: Decimal128('800.00')
}
```

The query filters apartments with a price greater than 100 and then groups the results by room_type and cancellation_policy.

Q9

```
> db.airbnb_listings.aggregate([
  { $match: { property_type: "Apartment", price: { $gt: 100 } } },
  {
    $group: {
      _id: { room_type: "$room_type", cancellation_policy: "$cancellation_policy" }
    }
  }
])
```

```
< {
  _id: {
    room_type: 'Private room',
    cancellation_policy: 'flexible'
  }
}
{
  _id: {
    room_type: 'Entire home/apt',
    cancellation_policy: 'moderate'
  }
}
{
  _id: {
    room_type: 'Entire home/apt',
    cancellation_policy: 'super_strict_60'
  }
}
{
  _id: {
    room_type: 'Shared room',
    cancellation_policy: 'moderate'
  }
}
{
  _id: {
    room_type: 'Entire home/apt',
    cancellation_policy: 'flexible'
  }
}
{
  _id: {
    room_type: 'Private room',
    cancellation_policy: 'super_strict_60'
  }
}
```

```

{
  _id: {
    room_type: 'Shared room',
    cancellation_policy: 'flexible'
  }
}
{
  _id: {
    room_type: 'Entire home/apt',
    cancellation_policy: 'super_strict_30'
  }
}
{
  _id: {
    room_type: 'Shared room',
    cancellation_policy: 'strict_14_with_grace_period'
  }
}
{
  _id: {
    room_type: 'Entire home/apt',
    cancellation_policy: 'strict_14_with_grace_period'
  }
}
}

}
{
  _id: {
    room_type: 'Private room',
    cancellation_policy: 'strict_14_with_grace_period'
  }
}
{
  _id: {
    room_type: 'Private room',
    cancellation_policy: 'moderate'
  }
}
}

```

Q10

The query filters all listings with property_type as "House", groups them by property_type and room_type, and then calculates the count of listings, the minimum value of minimum_nights, and the maximum value of maximum_nights for each group.


```
> db.airbnb_listings.aggregate([
  {
    "$match": {
      "property_type": "House"
    }
  },
  {
    "$group": {
      "_id": {
        "property_type": "$property_type",
        "room_type": "$room_type"
      },
      "count": { "$sum": 1 },
      "min_nights": { "$min": { "$toInt": "$minimum_nights" } },
      "max_nights": { "$max": { "$toInt": "$maximum_nights" } }
    }
  }
])
< {
  _id: {
    property_type: 'House',
    room_type: 'Entire home/apt'
  },
```

```
},  
  count: 330,  
  min_nights: 1,  
  max_nights: 1125  
}  
{  
  _id: {  
    property_type: 'House',  
    room_type: 'Shared room'  
  },  
  count: 9,  
  min_nights: 1,  
  max_nights: 1125  
}  
{  
  _id: {  
    property_type: 'House',  
    room_type: 'Private room'  
  },  
  count: 267,  
  min_nights: 1,  
  max_nights: 1125
```